

Name:
Student ID:
Signature:

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF MATHEMATICS AND STATISTICS

Term 2, 2025

MATH3311/MATH5335
Computational Mathematics for Finance

- (1) TIME ALLOWED – 2 HOURS
- (2) READING TIME – 10 MINUTES
- (3) THE EXAMINATION PAPER HAS 6 PAGES
- (4) TOTAL NUMBER OF QUESTIONS TO BE ANSWERED – 4
- (5) ANSWER ALL QUESTIONS
- (6) THE QUESTIONS ARE OF EQUAL VALUE
- (7) TOTAL NUMBER OF MARKS – 80
- (8) START EACH QUESTION ON A SEPARATE BOOKLET
- (9) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE
- (10) MATERIAL ALLOWED – ONE A4 PAGE WITH NOTES ON BOTH SIDES
(IT CAN BE PRINTED OR HANDWRITTEN), WHICH DOES NOT NEED
TO BE COLLECTED AT THE END OF THE EXAM

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [20 marks]

- i) Let $D \in \mathbb{R}^{m \times n}$, $B = D^T D$, and $\mathbf{b} \in \mathbb{R}^m$ with $m > n$. The *economic* singular value decomposition of D is $D = Y \Sigma V^T$ where $Y \in \mathbb{R}^{m \times n}$, $V \in \mathbb{R}^{n \times n}$ is orthogonal, and $\sigma_1 > \sigma_2 > \dots > \sigma_n > 0$ are the singular values.
- a) Write down the matrix Σ and its inverse Σ^+ .
 - b) Solve the linear system $D\mathbf{x} = \mathbf{b}$. Hint: Find $\mathbf{x}$ as a function of Y^T , Σ^+ , V and $\mathbf{b}$.
 - c) Show that the matrix B is positive semidefinite.
 - d) Show that the matrix B is positive definite.
- ii) Let $A \in \mathbb{R}^{n \times n}$ with a Cholesky factorisation $A = R^T R$ where $R = [r_{ij}] \in \mathbb{R}^{n \times n}$ is an upper triangular matrix with $r_{ii} > 0$ for $1 \leq i \leq n$.
- a) Show that $\det(A) > 0$.
 - b) Show that the Cholesky factorisation $A = R^T R$ may not be unique if the diagonal entries of R are not all positive.
 - c) Explain an efficient way to solve the linear system $RA\mathbf{x} = \mathbf{b}$.
 - d) What is the computational cost of solving $RA\mathbf{x} = \mathbf{b}$, given that computing the matrix R costs $n^3/3$ flops and solving a linear system by forward or backward substitution costs n^2 flops (each).
- Assume that $\kappa_2(R) = 2^n$ (the condition number of R in 2-norm).
- e) Verify that $\kappa_2(A) = 4^n$. Is A well-posed for any n ? Justify your answer.
 - f) In the linear system $A\mathbf{x} = \mathbf{b}$, the elements in A and $\mathbf{b}$ are known exactly. Estimate the relative error in computing $\mathbf{x}$.

2. [20 marks]

- i) Consider a system with the following floating point representation

$$\pm(1.d_1d_2d_3d_4)_2 \quad \text{where } d_1, d_2, d_3, d_4 \in \{0, 1\}.$$

- a) What is the machine epsilon (ϵ) for this system?
- b) By rounding the result of division to the nearest natural number multiple of ϵ , find the stored value of $x = 1 + 1.7/16$ on this floating point system.

- ii) Let
- f
- be a smooth function and
- $h > 0$
- . Consider the expression

$$f(x) - 2f(x+h) + f(x+2h) = h^2 f''(x) + h^3 f'''(x) + c h^4 f^{(4)}(x) + O(h^5).$$

- a) Use Taylor series expansions to find the constant c .
- b) Use the above expression to approximate $f''(x)$. What is the order of accuracy of this finite difference approximation?

- iii) Consider the following interest rate data

i	1	2	3	4	5
t_i (years)	0.25	0.5	1	2	3
$r(t_i)$	4.2 %	4.3 %	4.4 %	4.35 %	4.1 %

Table 1: Interest rate $r(t)$.

To estimate r over the time period $[0.25, 3]$ (years), an analyst used the following cubic spline interpolation

$$s(t) = \begin{cases} s_1(t) = a_1 + b_1(t - 0.25) + d_1(t - 0.25)^3, & t \in [0.25, 0.5], \\ s_2(t) = a_2 + b_2(t - 0.5) + c_2(t - 0.5)^2 + d_2(t - 0.5)^3, & t \in [0.5, 1], \\ s_3(t) = a_3 + b_3(t - 1) + c_3(t - 1)^2 + d_3(t - 1)^3, & t \in [1, 2], \\ s_4(t) = a_4 + b_4(t - 2) + c_4(t - 2)^2 + d_4(t - 2)^3, & t \in [2, 3]. \end{cases}$$

- a) Justify that the cubic spline s is of natural type.
- b) Use the given data and the properties of cubic spline interpolation to compute a_1, a_2, a_3, a_4 , and also to compute b_4 (in terms of b_3, c_3 and d_3).
- iv) Consider the 3-point Gauss–Legendre rule, which is exact for all polynomials of degree ≤ 5 ,

$$\int_{-1}^1 f(x) dx \approx w f(-\xi) + (2 - 2w) f(0) + w f(\xi),$$

where the weight w and the (positive) node ξ are such that $w\xi^2 = 1/3$ and $\xi^2 = 3/5$.

- a) Verify that the above 3-point Gauss–Legendre is NOT exact for some polynomials of degree ≤ 6 .
 - b) For smooth functions, if the above quadrature rule is scaled and applied over a small interval of length h , then the error is bounded by Mh^r for some positive constant M . What is the value of r ?
- v) Given the following set of financial data points (t_i, y_i) for $i = 1, 2, 3, 4$. The goal is to use the trigonometric linear function $\phi(\mathbf{x}; t) = x_1 + x_2 \cos(\pi t) + x_3 \sin(\pi t)$ to fit the given data set using the least square method, where $\mathbf{x} = (x_1, x_2, x_3)$.
- a) The residual column vector $\mathbf{r}(\mathbf{x})$ can be formulated in a matrix form as $\mathbf{r}(\mathbf{x}) = A\mathbf{x} - \mathbf{y}$. Write down the matrix A .
 - b) By using the least square method, the 2-norm of the residual $\|\mathbf{r}(\mathbf{z})\|_2$ minimize the $\|\mathbf{r}(\mathbf{x})\|_2$ for any $\mathbf{x} \in \mathbb{R}^3$. One way to find the estimator $\mathbf{z}$, is to solve the normal equation: $B\mathbf{z} = A^T\mathbf{y}$ where $B = A^T A$.
 - 1- When should this approach be avoided? Justify your answer.
 - 2- State an alternative approach (in one online) to compute $\mathbf{z}$.

3. [20 marks]

- i) Consider a real-valued random variable Y with a continuous, strictly decreasing, cumulative distribution function F , so that

$$\text{Prob}(Y \leq y) = F(y).$$

Let X be a uniformly distributed variable in the interval $[0, 1]$.

- a) Show that the random variable $F^{-1}(1 - X)$ has distribution with law F .
 - b) Suppose you have access to a reliable uniform random number generator that produces a sequence of independent samples $x_1, x_2, \dots$ for X . How can we use these samples to generate samples from Y ? What is likely to be the main practical difficulty with this approach?
- ii) The short-rate r_t (a type of interest rate) is assumed to follow the Ornstein–Uhlenbeck stochastic process

$$dr_t = \mu(\lambda - r_t)dt + \sigma dW_t, \quad t \in (0, T],$$

where $W_t = W(t)$ is a Wiener process, μ , λ and σ (volatility) are positive constants, and r_0 (the initial rate) is given. An explicit solution is given by

$$r_t = r_0 e^{-\mu t} + \lambda(1 - e^{-\mu t}) + \sigma \int_0^t e^{-\mu(t-s)} dW_s.$$

- a) Compute the expected value of r_t , $\mathbb{E}[r_t]$.
- b) Compute the variance of r_t , $\text{Var}[r_t]$.
- c) What is the distribution of r_t ?
- d) Describe how can you simulate N asset short-rate paths $(r_{t_0}^{(j)}, r_{t_1}^{(j)}, \dots, r_{t_R}^{(j)})$, $j = 1, 2, \dots, N$, at times $0 = t_0 < t_1 < t_2 < \dots < t_R = T$, where $r_{t_0} = r_0$. The time-points are equally spaced, i.e., with $\Delta t = T/R$, $t_n = n\Delta t$ for $n = 0, 1, 2, \dots, R$. Hint: You can use the discretization

$$r_{t_{r+1}} - r_{t_r} = \mu(\lambda - r_{t_r})\Delta t + \sigma\sqrt{\Delta t} Z_r,$$

where $Z_0, Z_1, \dots, Z_{R-1}$ are independent standard normal random variables.

- iii) In the analysis of compound interest, this type of recurrence relation appears:

$$G_n = 3G_{n-1} - 2G_{n-2}, \quad \text{for } n \geq 2, \text{ with } G_0 = 2 \text{ and } G_1 = 5.$$

- a) Estimate the number of needed flops to compute the n th term G_n using the above recurrence relation.
- b) One may show that $G_n = -1 + 3(2^n)$ for $n \geq 1$. State one numerical stability issue that might arise when using this formula in floating-point arithmetic, especially for large n .

4. [20 marks]

In this question we will use a European **cash-or-nothing call** option. This option pays a fixed amount $Q > 0$ at expiry T if the underlying asset price S at expiry exceeds the strike price K (that is, $S > K$); otherwise, it pays nothing.

Let $V(S, t)$ be the value of a European cash-or-nothing call option at time $t \in [0, T]$, when the underlying asset has price S . Under the Black–Scholes framework, the option price satisfies the PDE:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV, \quad \text{for } 0 < t < T \text{ and } 0 < S < \infty. \quad (1)$$

The risk-free interest rate r and the volatility σ are known positive constants.

- i) Sketch the graph of the pay-off function of the cash-or-nothing call option, i.e. the x -axis is the price of the underlying at expiry time T and the y -axis is the pay-off of the cash-or-nothing call option at expiry time.
- ii) Explain the terminal condition (i.e., $V(S, T)$) and the boundary conditions for the PDE governing this option.
- iii) Let $S_{\max} > K$ be the maximal price of the underlying used in the finite difference scheme and $t \in [0, T]$. Set up a grid $(S_j, t_n) = (j\Delta S, n\Delta t)$ for $j = 0, 1, 2, \dots, J$ and $n = 0, 1, 2, \dots, N$, where

$$\Delta S = \frac{S_{\max}}{J} \quad \text{and} \quad \Delta t = \frac{T}{N}.$$

Let V_j^n denote the approximate solution of $V(S_j, t_n)$ of (1) at the time t_n and underlying price S_j . Use the difference approximations,

$$\frac{\partial V}{\partial S}(S_j, t_n) \approx \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S}, \quad \frac{\partial V}{\partial t}(S_j, t_n) \approx \frac{V_j^n - V_j^{n-1}}{\Delta t},$$

and the second central difference approximation of $\frac{\partial^2 V}{\partial S^2}(S_j, t_n)$ to define the finite difference solution V_j^n .

- iv) Show that the finite difference equation has the form

$$V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n, \quad (2)$$

and find the coefficients α_j , β_j and γ_j .

- v) Use the boundary condition to define V_0^n and V_J^n .
- vi) Is the scheme in (2) explicit or implicit? Is it unconditionally stable?
- vii) Should one expect to lose accuracy in approximating $V(S, t)$ near $S = K$? If yes, explain why.
- viii) Say that the fixed amount $Q = j_0 \Delta S$ for some $1 < j_0 < J$. Use the finite difference scheme to compute $V_{j_0}^{N-2}$.

END OF EXAMINATION

Solution, 2025

Question 1

i)

a)

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \end{bmatrix} \quad \text{and} \quad \Sigma^+ = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} \end{bmatrix}$$

b) $D\mathbf{x} = \mathbf{b} \implies Y\Sigma V^T \mathbf{x} = \mathbf{b}$. Multiplying both sides by Y^T and using $Y^T Y = I$,

$$\Sigma V^T \mathbf{x} = Y^T \mathbf{b} \implies V^T \mathbf{x} = \Sigma^+ Y^T \mathbf{b}.$$

Now, multiply both sides by V and using $VV^T = I$, we get $\mathbf{x} = V\Sigma^+ Y^T \mathbf{b}$.

c) Let $\mathbf{z} \in \mathbb{R}^n$ be any vector. Then

$$\mathbf{z}^T B \mathbf{z} = \mathbf{z}^T D^T D \mathbf{z} = (D\mathbf{z})^T (D\mathbf{z}) = \|D\mathbf{z}\|_2^2 \geq 0.$$

Hence, B is positive semidefinite.

d) From part c), B is positive semidefinite. For positive definiteness, it is sufficient to show that if $D\mathbf{z} = \mathbf{0}$ then $\mathbf{z} = \mathbf{0}$. From part b), $D\mathbf{z} = \mathbf{0}$ implies $\mathbf{z} = V\Sigma^+ Y^T \mathbf{0} = \mathbf{0}$. Hence, B is positive definite.

ii) a)

$$\det(A) = \det(R^T R) = (\det(R))^2 = \left(\prod_{i=1}^n r_{ii} \right)^2 > 0 \text{ because } r_{ii} > 0 \text{ for all } 1 \leq i \leq n.$$

b) If we allow negative or arbitrary signs on the diagonal entries of R , then both R and $-R$ give the same product, that is, $R^T R = (-R)^T (-R)$. So the Cholesky factorization may not be unique if diagonal entries of R are not all positive.

c)

$$R A \mathbf{x} = \mathbf{b} \implies R(R^T R) \mathbf{x} = \mathbf{b}.$$

Let $\mathbf{y} = R^T R \mathbf{x}$ and $\mathbf{z} = R \mathbf{x}$. So, we need to solve the following three linear systems: $R \mathbf{y} = \mathbf{b}$ (backward substitution), $R^T \mathbf{z} = \mathbf{y}$ (forward substitution), and $R \mathbf{x} = \mathbf{b}$ (backward substitution).

d) The computational cost is: $\frac{n^3}{3} + 3n^2$ flops.

e) A is symmetric and positive definite, so $\kappa_2(A) = \lambda_n/\lambda_1$ where λ_n and λ_1 are the maximum and minimum eigenvalues of A , respectively. However, $\kappa_2(R) = \sqrt{\lambda_n}/\sqrt{\lambda_1}$, and so,

$$\kappa_2(A) = (\kappa_2(R))^2 = 4^n.$$

Since $\kappa_2(A)$ grows exponentially with n , the problem becomes increasingly ill-conditioned as n increases. Therefore, A is not well-posed for large n .

f) The relative error in computing $\mathbf{x}$ is approximately

$$\kappa_2(A)(\text{rel. error in } A + \text{rel. error in } b) = \kappa_2(A)(\epsilon + \epsilon) \approx 4^n (2\epsilon).$$

So the relative error grows with $\kappa_2(A)$.

Question 2

i) The floating point system is represented as: $\pm(1.d_1d_2d_3d_4)_2$.

a) The machine epsilon is the smallest number ε such that $1 + \varepsilon > 1$. In this system, the smallest value after 1 is when $d_1 = 0, d_2 = 0, d_3 = 0, d_4 = 1$, so:

$$\varepsilon = (1.0001)_2 - 1 = (0.0001)_2 = 2^{-4} = \frac{1}{16}$$

b) $x = 1 + \frac{1.7}{16} = 1 + 1.7\varepsilon$. So the stored value is $1 + 2\varepsilon = 1 + 1/8 = 9/8$.

ii)

a) Using Taylor series expansions:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) + O(h^5)$$

$$f(x+2h) = f(x) + 2hf'(x) + \frac{4h^2}{2}f''(x) + \frac{8h^3}{6}f'''(x) + \frac{16h^4}{24}f^{(4)}(x) + O(h^5)$$

Then,

$$f(x) - 2f(x+h) + f(x+2h) = h^2f''(x) + h^3f'''(x) + \frac{7}{12}h^4f^{(4)}(x) + O(h^5) \implies c = \frac{7}{12}$$

b) From the above expansion,

$$f''(x) = \frac{f(x) - 2f(x+h) + f(x+2h)}{h^2} - hf'''(x) - \frac{7}{12}h^2f^{(4)}(x) + O(h^3),$$

and so, $f''(x)$ can be approximated as:

$$f''(x) \approx \frac{f(x) - 2f(x+h) + f(x+2h)}{h^2}$$

It is $O(h)$ accurate.

iii)

a) A natural cubic spline has zero second derivative at endpoints. The cubic spline s is of natural type if $s''(0.25) = s''(3) = 0$, that is, $s_1''(0.25) = s_4''(3) = 0$.

$$s_1''(t) = 6d_1(t - 0.25) \quad \text{and} \quad s_4''(t) = -6d_4 + 6d_4(t - 2) \Rightarrow s_1''(0.25) = 0 \quad \text{and} \quad s_4''(3) = 0.$$

b) From the table:

$$a_1 = r(0.25) = 0.042, \quad a_2 = r(0.5) = 0.043, \quad a_3 = r(1) = 0.044, \quad a_4 = r(2) = 0.0435$$

To find b_4 , we differentiate s_3 and s_4 and match derivatives at $t = 2$:

$$s_3'(t) = b_3 + 2c_3(t - 1) + 3d_3(t - 1)^2, \quad s_4'(t) = b_4 - 6d_4(t - 2) + 3d_4(t - 2)^2$$

At $t = 2$:

$$s_4'(2) = s_3'(2) \Rightarrow b_4 = b_3 + 2c_3 + 3d_3.$$

iv)

a) Choose $f(x) = x^6$, then

$$\int_{-1}^1 x^6 dx = \frac{2}{7},$$

and

$$wf(-\xi) + (2 - 2w)f(0) + wf(\xi) = 2w\xi^6 = 2(w\xi^2)\xi^4 = 2(1/3)(3/5)^2 = 6/25.$$

Therefore,

$$\int_{-1}^1 f(x) dx \neq wf(-\xi) + (2 - 2w)f(0) + wf(\xi)$$

when $f(x) = x^6$, and so, the given 3-point Gauss–Legendre is NOT exact for some polynomials of degree ≤ 6 .

b) Since the 3-point Gauss–Legendre of degree ≤ 5 only, $r = 5 + 1 = 6$.

v)

a) The matrix A is:

$$A = \begin{bmatrix} 1 & \cos(\pi t_1) & \sin(\pi t_1) \\ 1 & \cos(\pi t_2) & \sin(\pi t_2) \\ 1 & \cos(\pi t_3) & \sin(\pi t_3) \\ 1 & \cos(\pi t_4) & \sin(\pi t_4) \end{bmatrix}$$

b)

1- This approach should be avoided when the matrix A is ill-conditioned, that is, $\kappa_2(A)$ is large.

2- For an alternative approach, one may use QR decomposition or SVD which are more stable numerically.

Question 3

i)

a) Let $Z = F^{-1}(1 - X)$. We compute the CDF of Z :

$$\begin{aligned}\text{Prob}(Z \leq y) &= \text{Prob}(F^{-1}(1 - X) \leq y) = \text{Prob}(1 - X \leq F(y)) \quad (\text{since } F \text{ is decreasing}) \\ &= \text{Prob}(X \geq 1 - F(y)) = 1 - \text{Prob}(X < 1 - F(y)) = 1 - (1 - F(y)) = F(y).\end{aligned}$$

Hence, Z has distribution F .

b) Compute $y_i = F^{-1}(1 - x_i)$ for each i . Then $y_1, y_2, \dots$ are samples from the distribution of Y . The main practical difficulty is that the function F^{-1} may not be available in closed form or may be computationally expensive to evaluate.

ii)

Given the SDE:

$$dr_t = \mu(\lambda - r_t)dt + \sigma dW_t, \quad r_0 \text{ given,}$$

the explicit solution is:

$$r_t = r_0 e^{-\mu t} + \lambda(1 - e^{-\mu t}) + \sigma \int_0^t e^{-\mu(t-s)} dW_s$$

a) Let $\Delta s = t/N$ and $s_n = n\Delta s$.

$$\int_0^t e^{-\mu(t-s)} dW_s = \lim_{\Delta s \rightarrow 0} \sum_{n=0}^{N-1} e^{-\mu(t-s_n)} (W(s_{n+1}) - W(s_n)).$$

To find the expected value,

$$\begin{aligned}\mathbb{E} \left[\int_0^t e^{-\mu(t-s)} dW_s(s) \right] &= \lim_{\Delta s \rightarrow 0} \sum_{n=0}^{N-1} \mathbb{E} \left[e^{-\mu(t-s_n)} (W(s_{n+1}) - W(s_n)) \right] \\ &= \lim_{\Delta s \rightarrow 0} \sum_{n=0}^{N-1} e^{-\mu(t-s_n)} \mathbb{E} [W(s_{n+1}) - W(s_n)] = 0\end{aligned}$$

and so, $\mathbb{E}[r_t] = r_0 e^{-\mu t} + \lambda(1 - e^{-\mu t})$.

b) To find the variance,

$$\begin{aligned}\mathbb{V}ar \left[\int_0^t e^{-\mu(t-s)} dW_s(s) \right] &= \lim_{\Delta s \rightarrow 0} \sum_{n=0}^{N-1} \mathbb{V}ar \left[e^{-\mu(t-s_n)} (W(s_{n+1}) - W(s_n)) \right] \\ &= \lim_{\Delta s \rightarrow 0} \sum_{n=0}^{N-1} e^{-2\mu(t-s_n)} \Delta s = \int_0^t e^{-2\mu(t-s)} ds\end{aligned}$$

and so,

$$\mathbb{V}ar[r_t] = \sigma^2 \int_0^t e^{-2\mu(t-s)} ds = \frac{\sigma^2}{2\mu} (1 - e^{-2\mu t}).$$

c) Since

$$W_t \sim \mathcal{N}(0, t), \quad \mathbb{E}[r_t] = r_0 e^{-\mu t} + \lambda(1 - e^{-\mu t}), \quad \text{Var}[r_t] = \frac{\sigma^2}{2\mu}(1 - e^{-2\mu t}),$$

then

$$r_t \sim \mathcal{N}\left(r_0 e^{-\mu t} + \lambda(1 - e^{-\mu t}), \frac{\sigma^2}{2\mu}(1 - e^{-2\mu t})\right).$$

d) We discretize the process as:

$$r_{t_{r+1}} = r_{t_r} + \mu(\lambda - r_{t_r})\Delta t + \sigma\sqrt{\Delta t}Z_r,$$

where $Z_r \sim \mathcal{N}(0, 1)$ are i.i.d. normal random variables.

Algorithm:

- Initialize r_0 for all N paths.
- For each time step t_r , $r = 0, 1, \dots, R - 1$, and for each path $j = 1, 2, \dots, N$:

$$r_{t_{r+1}}^{(j)} = r_{t_r}^{(j)} + \mu(\lambda - r_{t_r}^{(j)})\Delta t + \sigma\sqrt{\Delta t}Z_r^{(j)}.$$

iii) Given the recurrence relation:

$$G_n = 3G_{n-1} - 2G_{n-2}, \quad G_0 = 2, \quad G_1 = 5.$$

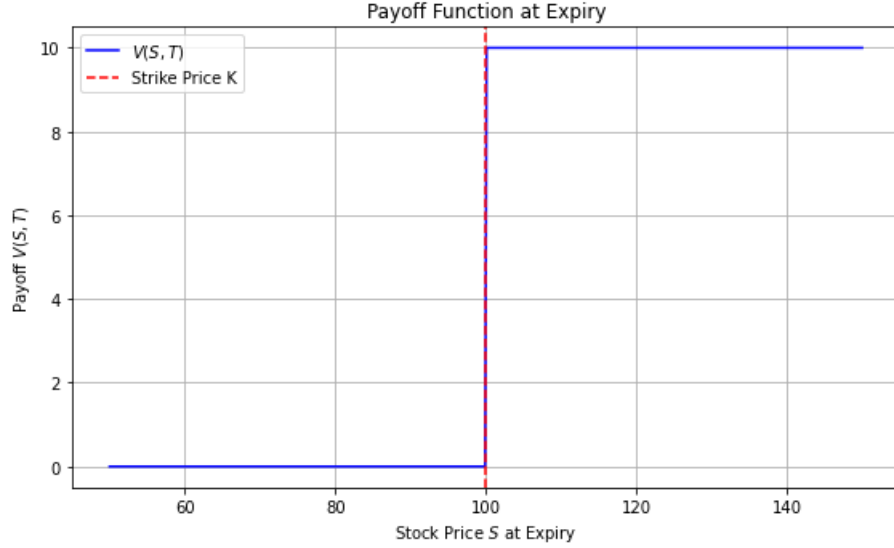
a) To compute G_n iteratively, each step requires 3 operations (starting from G_2), and so, the total number of operations is $3(n - 1) \approx 3n$.

b) When n is large, 2^n becomes extremely large, and overflow may occur due to limitations in representing large numbers.

Question 4

i) The payoff function at expiry is a step function with a discontinuity at $S = K$.

In the figure below, $K = 100$ and $Q = 10$.



ii)

Terminal condition at time T :

$$V(S, T) = \begin{cases} 0, & S \leq K \\ Q, & S > K \end{cases}$$

Boundary conditions:

- As $S \rightarrow 0$, the option will not be exercised, so $V(0, t) = 0$.
- As $S \rightarrow \infty$, the option will be exercised with certainty $V(S, t) \rightarrow Qe^{-r(T-t)}$.

iii)

Finite difference approximations:

$$\frac{\partial V}{\partial t}(S_j, t_n) \approx \frac{V_j^n - V_j^{n-1}}{\Delta t}, \quad \frac{\partial V}{\partial S}(S_j, t_n) \approx \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S}, \quad \frac{\partial^2 V}{\partial S^2}(S_j, t_n) \approx \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2}.$$

Substitute the approximations into the PDE:

$$\frac{V_j^n - V_j^{n-1}}{\Delta t} + rS_j \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2} = rV_j^n.$$

iv) The finite different solution can be rewritten as:

$$V_j^{n-1} = \alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n,$$

where

$$\alpha_j = \frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} - \frac{rS_j \Delta t}{2\Delta S}, \quad \beta_j = 1 - \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2} - r\Delta t, \quad \gamma_j = \frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} + \frac{rS_j \Delta t}{2\Delta S}.$$

v) Boundary conditions in the scheme: for all n ,

- At $S = 0$ ($j = 0$): $V_0^n = 0$ and at $S = S_{\max}$ ($j = J$): $V_J^n = Qe^{-r(T-t_n)}$.

vi)

The scheme is explicit.

It is conditionally stable and one may require $\Delta t \leq C(\Delta S)^2$ for some positive constant C to guarantee the stability.

vii) Yes, loss of accuracy is expected near $S = K$ because V has discontinuity at $S = K$, so the solution is not smooth.

viii) From the terminal condition at time T , $V_i^N = 0$ for $i \leq j_0$ and $V_i^N = Q$ for $i > j_0$. So, from the finite difference scheme

$$\begin{aligned} V_{j_0}^{N-1} &= \alpha_{j_0} V_{j_0-1}^N + \beta_{j_0} V_{j_0}^N + \gamma_{j_0} V_{j_0+1}^N = \gamma_{j_0} Q \\ V_{j_0+1}^{N-1} &= \alpha_{j_0+1} V_{j_0}^N + \beta_{j_0+1} V_{j_0+1}^N + \gamma_{j_0+1} V_{j_0+2}^N = \beta_{j_0+1} Q + \gamma_{j_0+1} Q \\ V_{j_0-1}^{N-1} &= \alpha_{j_0-1} V_{j_0-2}^N + \beta_{j_0-1} V_{j_0-1}^N + \gamma_{j_0-1} V_{j_0}^N = 0 \end{aligned}$$

Then,

$$V_{j_0}^{N-2} = \alpha_{j_0} V_{j_0-1}^{N-1} + \beta_{j_0} V_{j_0}^{N-1} + \gamma_{j_0} V_{j_0+1}^{N-1} = \beta_{j_0} \gamma_{j_0} Q + \gamma_{j_0} (\beta_{j_0+1} + \gamma_{j_0+1}) Q$$